Lunar Surface Analysis via XRF

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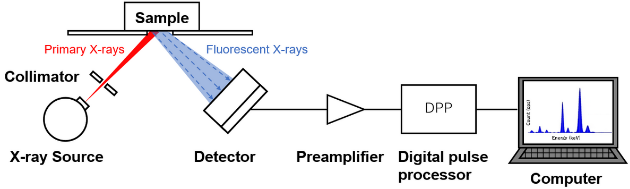
Introduction

- **Goal:** Identification and quantification of chemical elements on the lunar surface.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of `multiel-spectra` for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, different for each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.



Project Phases

The analysis focuses on determining the chemical elements within the XRF spectra through three interconnected computational tasks:

1. Classification
2. Regression
3. Denoising

Synthetic Data Generation

To train models, we developed a Generator that simulates physical variations in XRF spectra.

- **Composition:** Samples contain **0 to 5** elements randomly selected from **41** possible candidates.
- **Stochasticity:** To ensure model robustness, each spectrum is generated by randomly varying operating conditions and noise levels simulating real-world measurement variability.

Classification

The classification task focus on identifying the elemental composition by analyzing spectral signatures.

Analytical stages

- **Multi-label classification:** across a pool of 41 potential elements.
- **Multi-class classification:** across 5 classes (1 to 5 elements) defined by the number of elements present in the sample.

Preprocessing Strategies and Architectures

Preprocessing Scaling

Tested to handle high dynamic range in spectral counts:

- **Standard:** 0 mean and 1 variance.
- **MinMax:** Rescaling to $[0, 1]$ range.
- **Log-MinMax:** Logarithmic transformation to compress peaks.

Model Architectures

Machine Learning:

- Random Forest, AdaBoost, XGBoost

Deep Learning:

- MLP, CNN, Transformer

Results: Example Multi-label Classification

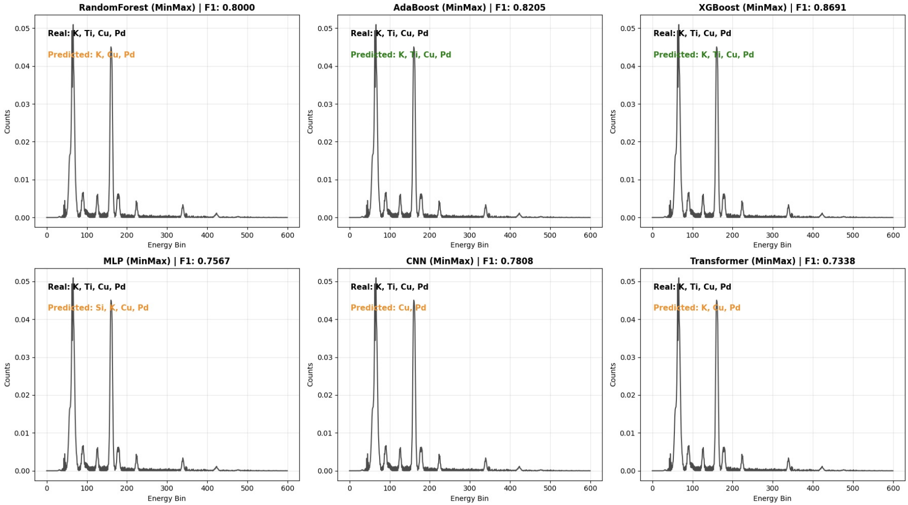


Figure: Predicted vs. real elemental composition for a sample

Results Multi-label Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
Standard	XGBoost	0.8691	0.9593	0.8237
MinMax	XGBoost	0.8691	0.9589	0.8243
LogMinMax	XGBoost	0.8691	0.9589	0.8243
MinMax	AdaBoost	0.8205	0.9151	0.7821
LogMinMax	AdaBoost	0.8203	0.9088	0.7830
Standard	AdaBoost	0.8142	0.9049	0.7770
LogMinMax	RandomForest	0.8004	0.9618	0.7314
MinMax	RandomForest	0.8000	0.9600	0.7305
Standard	RandomForest	0.7995	0.9848	0.7292
LogMinMax	CNN	0.7979	0.9719	0.7267
Standard	CNN	0.7910	0.9597	0.7203
MinMax	CNN	0.7808	0.9567	0.7062
MinMax	MLP	0.7567	0.9419	0.6802
LogMinMax	MLP	0.7542	0.9494	0.6751
MinMax	Transformer	0.7338	0.9441	0.6540
Standard	MLP	0.7320	0.9351	0.6415
Standard	Transformer	0.7173	0.8914	0.6331
LogMinMax	Transformer	0.7141	0.9299	0.6295

Figure: Multi-label classification results ranked by F1-Score.

Results Multi-class Classification

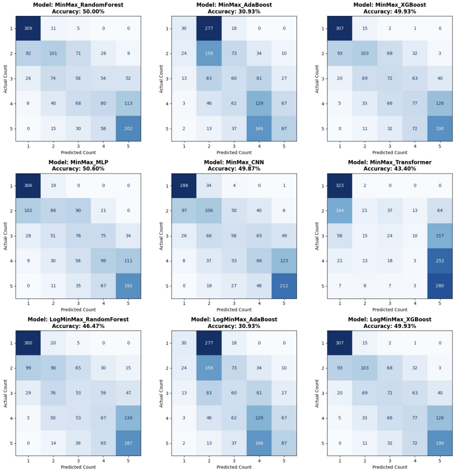


Figure: Confusion matrices for multi-class classification.

Results: Example Multi-class Classification

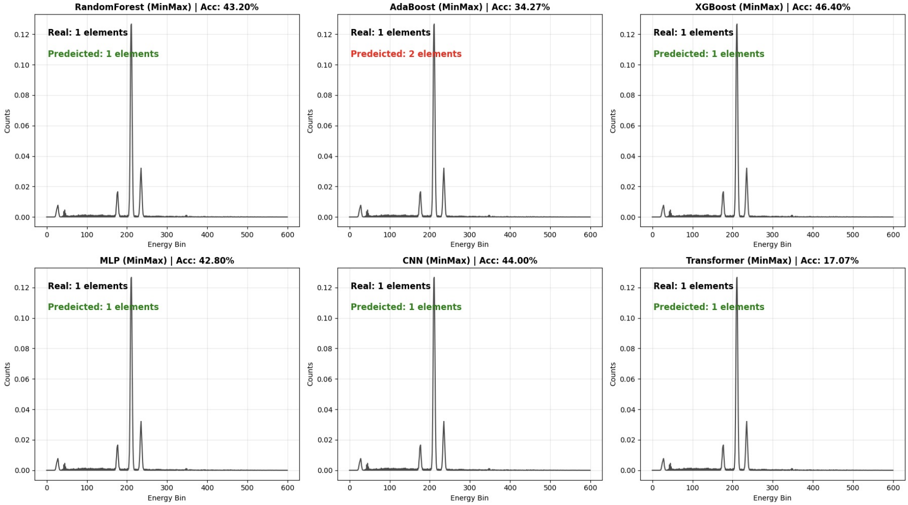


Figure: Predicted vs. real number of elements.

Results Multi-class Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
LogMinMax	XGBoost	0.4473	0.4411	0.4653
MinMax	XGBoost	0.4467	0.4402	0.4640
Standard	RandomForest	0.4288	0.4313	0.4440
Standard	XGBoost	0.4240	0.4203	0.4413
MinMax	CNN	0.4207	0.4330	0.4400
LogMinMax	CNN	0.4185	0.4230	0.4307
MinMax	RandomForest	0.4137	0.4148	0.4320
MinMax	MLP	0.4023	0.4007	0.4280
LogMinMax	RandomForest	0.3820	0.3780	0.4027
LogMinMax	MLP	0.3686	0.3786	0.4133
Standard	CNN	0.3650	0.3579	0.3853
Standard	AdaBoost	0.3327	0.3489	0.3427
MinMax	AdaBoost	0.3327	0.3489	0.3427
LogMinMax	AdaBoost	0.3327	0.3489	0.3427
Standard	MLP	0.2388	0.2694	0.3013
Standard	Transformer	0.0802	0.0490	0.2213
LogMinMax	Transformer	0.0776	0.0472	0.2173
MinMax	Transformer	0.0498	0.0291	0.1707

Figure: Multi-class classification results ranked by F1-Score.

Regression Task

The regression task focuses on **quantifying** the elemental composition by predicting concentration values from spectral data.

Problem formulation

- **Input:** XRF spectrum — 600 energy bins (0–30 keV)
- **Output:** 41 elemental concentrations, non-negative and summing to 1
- **Evaluation:** Masked MAE and R^2 computed only on *active* elements (ground truth > 0)

Key challenge

The output lies on the probability simplex and has **sparse structure** (only 1–5 of 41 elements active per sample) — standard regression metrics are deceptive, only performance on active elements counts.

Baseline Models & Initial Results

Models compared

- **CNNRegressor:** Conv1D + Global Average Pool + dense head
- **ResNetRegressor:** residual blocks
- **PLS:** Partial Least Squares (20 components)
- **Ridge:** L2-regularised, one model per element

Initial results (5k samples)

Model	MAE ↓	R^2 ↑
CNN	0.259	-1.17
ResNet	0.256	-1.12
PLS	0.227	-0.74
Ridge	0.210	-0.47

All R^2 values are **negative** — models worse than mean predictor.

Why CNN Failed: Architecture Issues

Problems with CNNRegressor V1

- **Global Average Pool** collapses all spatial info — cannot distinguish *where* peaks are
- **MaxPool** discards amplitude — concentration $\propto$ peak area, not maximum
- **ReLU + normalise** output collapses to uniform predictions when uncertain

CNNRegressorV2 fixes

- **AvgPool** instead of MaxPool — preserves amplitude
- **No GAP** — retain 15 spatial positions before flatten
- **Softmax** output — stable normalisation

V2 result (10k samples)

$\text{MAE} = 0.212$, $R^2 = -0.477$

Still negative — CNN cannot learn peak positions from data alone at this scale.

Physics-Informed Peak Features

Key insight: XRF emission line energies are *physical constants* — we know exactly which spectral bin each element's K-α, K-β, L-α peak appears in.

Multi-line peak feature extraction

For each of the 41 elements, integrate counts in $\pm$5-bin windows around all in-range emission lines $\Rightarrow$ **129 non-redundant features** encoding direct physical knowledge.

Also computed:

- **Ratio features:** $r_i = p_i / \sum_j p_j$ — directly proportional to concentration
- **Classifier probs:** 41 sigmoid outputs from CNNClassifier — tells which elements are present

Results (10k samples)

Features	MAE	R^2
Peak integrals (129)	0.209	−0.41
+ ratios	0.206	−0.34
+ clf probs (299)	0.206	−0.34

Per-Element Decomposition

Key insight: instead of one model for all 41 outputs, train one model *per element* — each sees only the peaks relevant to it and its overlapping neighbours.

Overlap-aware feature set per element

For element i: collect peak integrals of all elements whose emission lines fall within ± 10 bins of any of i's lines. Fe sees Co, Ni, Cr, O, Na, etc. — its known physical overlaps.

- 41 Ridge regressors, each on element-specific features
- Classifier probabilities concatenated to every sub-model
- Outputs clipped ≥ 0 and renormalised

Result (10k samples)

$$\text{MAE} = 0.195,\ \text{R}^2 = -0.183$$

Classifier features provide a **massive jump**: Ridge alone gives -0.44.

Element Transformer

Key insight: each element is a **token** — self-attention lets elements learn inter-element dependencies (overlap, sum-to-1, co-occurrence).

Token embedding (per element)

- Peak integrals (abs + ratio $r_i = p_i / \sum_j p_j$) for own lines
- Normalised atomic number $Z/92$, primary bin index
- Classifier probability for that element

Architecture:

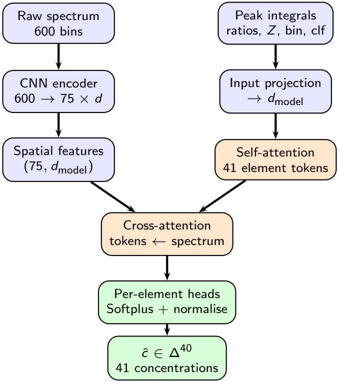
- 41 tokens → d_model=128 projection
- 3 Transformer encoder layers, 8 heads
- Per-element output head → Softplus normalisation

Result (10k samples)

$$\text{MAE} = 0.191, \ \text{R}^2 = -0.135$$

Self-attention captures which elements co-occur and how their signals interact.

Best Model: Transformer V2 Architecture



MAE $= 0.190$, $\mathbf{R^2} = -0.114$ (10k samples, mixup + curriculum + hard mask)

Training Curve & Per-Element Accuracy

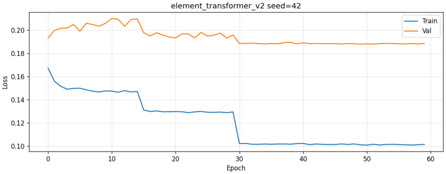


Figure: Training vs validation MAE loss.

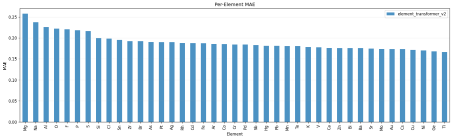


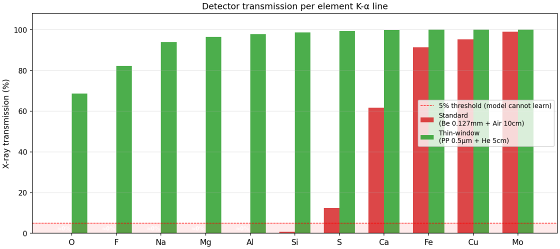
Figure: Per-element MAE on test set.

Results Summary & Model Progression

Model	MAE ↓	R^2 ↑
CNN V1 (Global Avg Pool)	0.259	−1.17
CNN V2 (AvgPool + spatial head)	0.212	−0.477
Ridge (full spectrum)	0.210	−0.472
MLP on 299 features (ratio + clf)	0.206	−0.343
Per-element Ridge + clf probs	0.195	−0.183
Element Transformer	0.191	−0.135
Transformer V2 (cross-attn)	**0.190**	**−0.114**

$R^2 > 0$ requires ∼12% further MSE reduction — achievable with ≥20k samples.

Physical Bottleneck: Detector Transmission



X-ray attenuation law

$$T = \prod_i e^{-\frac{\mu}{\rho}\big|_i \rho_i d_i}$$

μ/ρ: mass att. coeff. (NIST, cm^2/g), ρ: density, d: thickness

O K-α, Be 0.127 mm: $e^{-8273\times1.85\times0.013} \approx e^{-194} \approx 0$

O K-α, PP 0.5 μm + He: $e^{-9961\times0.86\times5\times10^{-5}} \approx 0.65$

XRF Denoising

- **The Problem:** XRF spectra are often corrupted by noise and background artifacts.
- **The Impact:** Noise can obscure low-intensity peaks, leading to incorrect elemental identification or inaccurate quantification.
- **The Objective:** "Chemical signal recovery": mapping a noisy raw input spectrum to its clean, noise-free ground-truth counterpart.

Dataset Generation

To synthesize the required (X, Y) training pairs, we developed a high-performance generator:

- **Decisional Logic (`n_counts`):** Every pair (X, Y) is generated by fixing a specific physical state θ and synthesizing two versions via the `multiel_spectra` engine:
  - **Ground Truth (Y):** `n_counts = 1` (minimal noise, simulator floor).
  - **Noisy Observation (X):** `n_counts` $\in [10000, 30000]$ (stochastic degradation).
- **Generic Learning:** This variety prevents overfitting to a fixed noise profile, forcing the model to learn pure signal recovery.
- **Dataset Size:** 2000 paired samples; split 70/15/15 (train/val/test).

Preprocessing Strategies

XRF spectra present a unique **"Peak vs. Baseline"** challenge: peaks can be orders of magnitude larger than the background.

- **Standard Scaling:** Normalizes by mean and standard deviation; preserves the natural peak-to-background ratio and proved most robust across all architectures.
- **MinMax Scaling:** Normalizes to [0,1]; often suppresses trace signals in the presence of dominant peaks.
- **LogMinMaxScaler:** Applies $\log(1 + x)$ followed by MinMax scaling; theoretically compresses dynamic range to protect low-intensity peaks, but did not consistently outperform Standard scaling in practice.

Model Architectures (1/2)

We evaluated 8 diverse architectures to find the best structural fit for 1D spectral data:

- **CNN (1D):** Standard convolutional autoencoder. Uses local kernels to detect and "smooth" peak shapes.
- **UNet1D:** Deep encoder-decoder with skip-connections. Excellent for preserving high-resolution peak positions while suppressing noise.
- **Transformer:** Treats spectral channels as a token sequence. Applies self-attention across the full spectrum to model relationships between distant energy regions.
- **Conformer:** Combines local convolutions (shape detection) with global attention (peak relationships).

Model Architectures (2/2)

- **ResNet1D:** Utilizes residual blocks to allow deeper feature extraction without losing original signal information.
- **LSTM:** Recurrent autoencoder. Processes the spectrum as a sequence of energy steps.
- **MLP:** Fully connected autoencoder.
- **Linear:** Linear bottleneck.

Training Objective Design

We systematically evaluated **4 loss strategies** defined by two orthogonal choices:

	No Sparsity	**+ L1 on Background**
Global MSE	Standard MSE	Standard MSE + L1
Masked MSE	Masked MSE	Masked MSE + L1

- **Global vs. Masked:** Whether the MSE loss is computed over the full spectrum or only over peak regions.
- **L1 Sparsity:** Penalizes any non-zero prediction in flat background areas, actively suppressing residual artifacts that MSE alone tends to leave behind.

Training Stability: Standard MSE

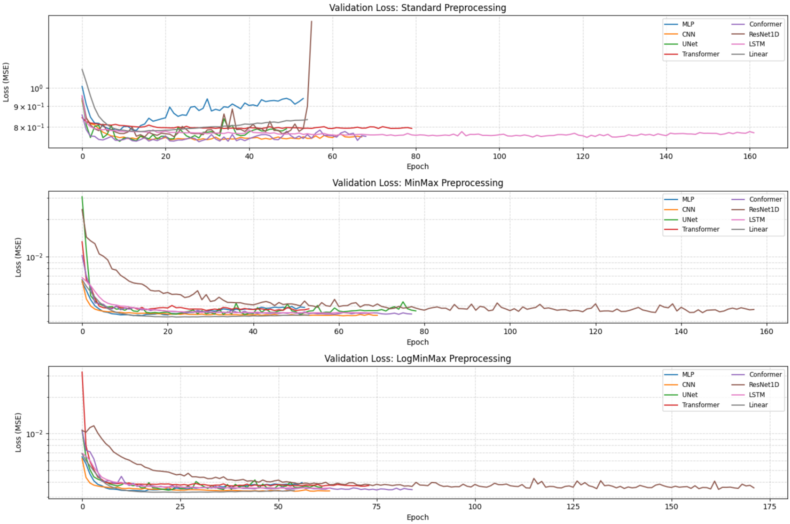


Figure: Validation loss convergence for all models trained with Standard MSE loss.

Training Stability: Standard MSE + L1

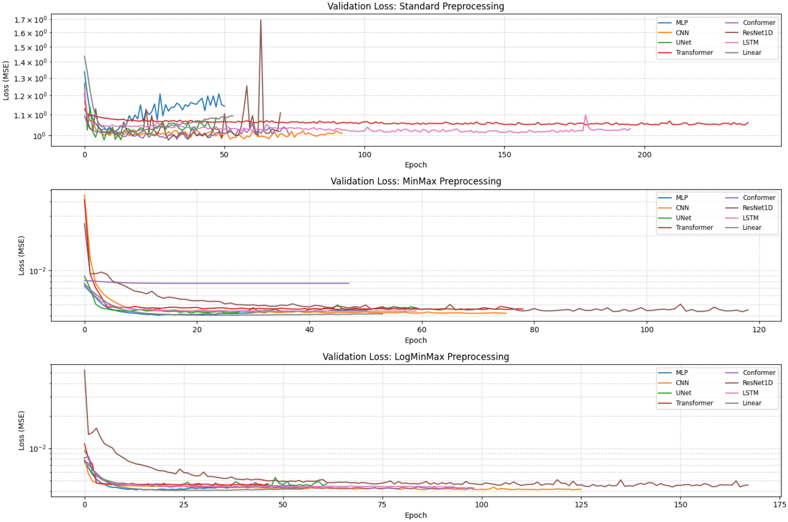


Figure: Validation loss convergence for all models trained with Standard MSE + L1 loss.

Training Stability: Masked MSE

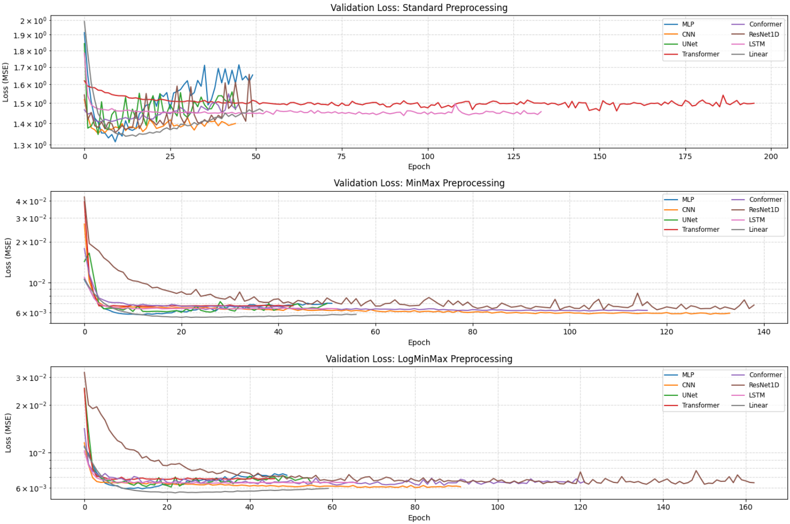


Figure: Validation loss convergence for all models trained with Masked MSE loss.

Training Stability: Masked MSE + L1

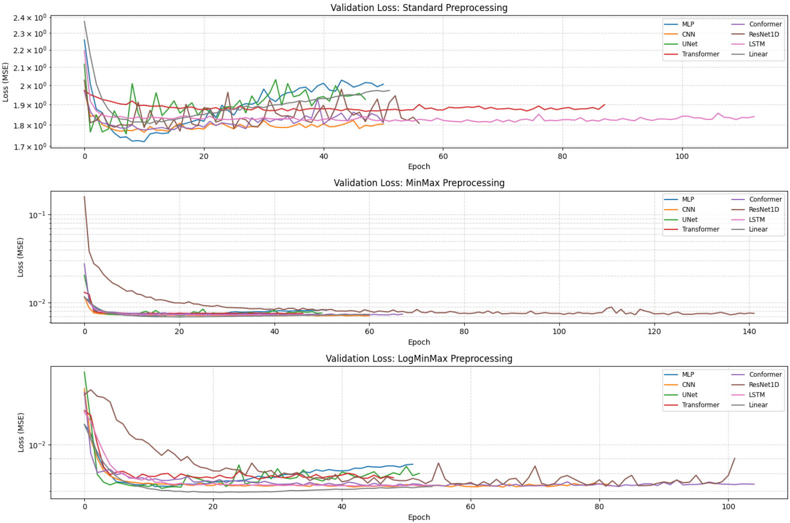


Figure: Validation loss convergence for all models trained with Masked MSE + L1 loss.

Results: Standard MSE

Architectures sorted by Standard scaler MSE. Bold = best per column.

MSE ($\times 10^{-5}$)

Arch.	**Std**	**LogMM**	**MM**
Conformer	**3.52**	4.02	4.18
UNet1D	3.68	3.96	4.42
LSTM	3.71	3.96	**3.91**
CNN	4.05	4.35	4.30
MLP	4.18	4.50	5.16
ResNet1D	4.31	4.31	4.30
Transformer	4.37	**4.07**	4.09
Linear	4.45	4.80	5.15

MAE ($\times 10^{-3}$)

Arch.	**Std**	**LogMM**	**MM**
Conformer	**1.16**	1.47	1.52
UNet1D	1.20	1.40	1.39
LSTM	1.33	1.36	1.35
CNN	1.23	1.42	1.40
MLP	1.68	1.62	1.71
ResNet1D	1.26	1.52	1.67
Transformer	1.32	**1.21**	**1.14**
Linear	1.64	1.63	1.73

Results: Standard MSE + L1

Architectures sorted by Standard scaler MSE. Bold = best per column.

MSE ($\times 10^{-5}$)			
Arch.	**Std**	**LogMM**	**MM**
CNN	**3.58**	**4.38**	4.71
Conformer	3.93	4.45	11.70
LSTM	4.13	4.40	**4.24**
Linear	4.19	4.77	5.12
Transformer	4.36	4.54	4.55
MLP	4.37	4.92	5.21
UNet1D	4.88	4.63	4.46
ResNet1D	4.94	4.74	4.86

MAE ($\times 10^{-3}$)			
Arch.	**Std**	**LogMM**	**MM**
CNN	1.25	1.43	1.56
Conformer	1.22	1.30	2.40
LSTM	1.34	**1.27**	**1.29**
Linear	1.66	1.64	1.71
Transformer	1.44	1.40	1.35
MLP	1.72	1.68	1.70
UNet1D	**1.20**	1.39	1.38
ResNet1D	1.32	1.51	1.56

Results: Masked MSE

Architectures sorted by Standard scaler MSE. Bold = best per column.

MSE ($\times 10^{-5}$)

Arch.	**Std**	**LogMM**	**MM**
Conformer	**6.88**	7.27	7.54
UNet1D	7.01	7.29	7.62
Transformer	7.20	7.77	7.72
CNN	7.57	**6.89**	**6.94**
LSTM	7.59	7.45	7.47
Linear	7.60	7.25	7.39
ResNet1D	8.00	7.78	7.89
MLP	8.48	8.57	7.92

MAE ($\times 10^{-3}$)

Arch.	**Std**	**LogMM**	**MM**
Conformer	1.93	1.97	2.39
UNet1D	**1.92**	2.12	2.15
Transformer	2.09	**1.85**	**1.95**
CNN	2.13	2.01	2.18
LSTM	2.23	2.10	2.01
Linear	2.48	2.30	2.39
ResNet1D	2.17	2.38	2.35
MLP	2.54	2.50	2.46

Results: Masked MSE + L1

Architectures sorted by Standard scaler MSE. Bold = best per column.

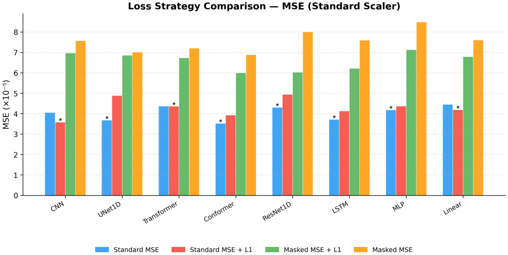
MSE ($\times 10^{-5}$)

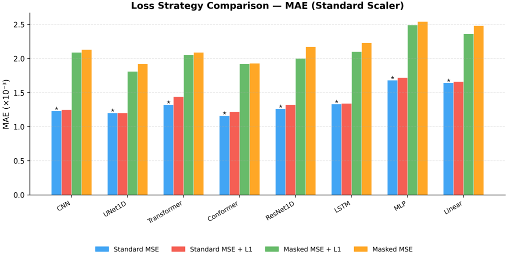
Arch.	**Std**	**LogMM**	**MM**
Conformer	**5.99**	6.16	6.75
ResNet1D	6.03	6.93	6.95
LSTM	6.21	6.74	**6.55**
Transformer	6.73	6.69	6.74
Linear	6.79	6.92	6.96
UNet1D	6.86	**6.50**	7.17
CNN	6.97	6.72	7.08
MLP	7.13	7.35	7.41

MAE ($\times 10^{-3}$)

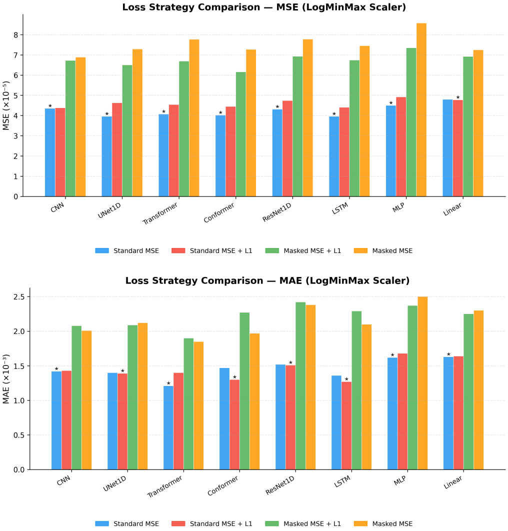
Arch.	**Std**	**LogMM**	**MM**
Conformer	1.92	2.27	2.08
ResNet1D	2.00	2.42	2.42
LSTM	2.10	2.29	**2.07**
Transformer	2.05	**1.90**	2.19
Linear	2.36	2.25	2.30
UNet1D	**1.81**	2.09	2.25
CNN	2.09	2.08	2.31
MLP	2.49	2.37	2.45

Loss Strategy Comparison: Standard Scaler

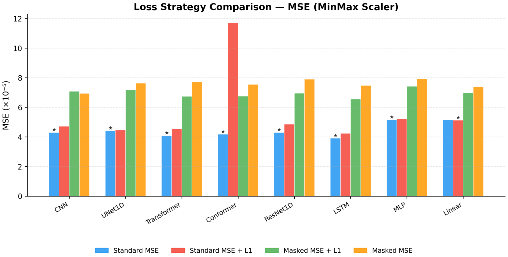


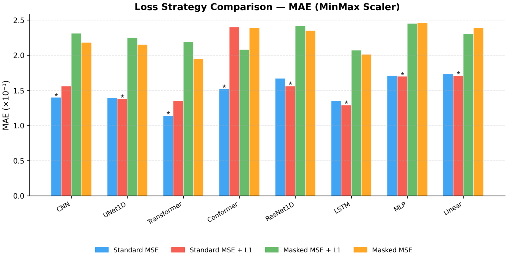


Loss Strategy Comparison: LogMinMax Scaler



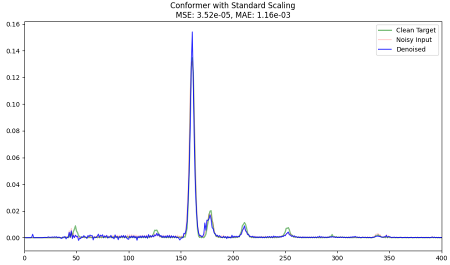
Loss Strategy Comparison: MinMax Scaler



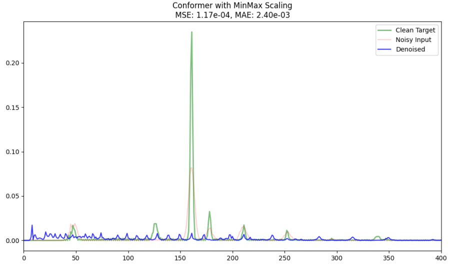


Reconstruction Showcase: Global Best vs. Worst

Best: Conformer, Standard MSE, Standard scaler
MSE: 3.52×10^{-5}



Worst: Conformer, MSE + L1, MinMax scaler
MSE: 11.70×10^{-5}



The Denoising Champion: Conformer + Standard MSE

- **Best MSE:** 3.52×10^{-5}, lowest across all architectures and strategies.
- **Design:** Local convolutions for peak shapes; global attention for long-range spectral structure.
- **Standard Scaler:** Preserves the peak-to-background ratio, ideal for the Conformer's dual inductive biases.
- **No Masking:** Global MSE suffices; attention naturally focuses on informative regions.

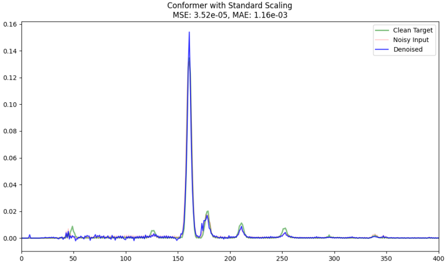


Figure: Best reconstruction by the Conformer (Standard MSE).